

统计信号处理基础

第八章

贝叶斯估计 (二)

清华大学电子工程系

李洪 教授

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内容概要

- 一、矢量参数贝叶斯估计
- 二、贝叶斯风险
- 三、线性贝叶斯估计
- 四、序贯LMMSE
- 五、小结



一、矢量参数贝叶斯估计

经典估计MSE:

$$\begin{aligned}\text{mse}(\hat{\theta}) &= E\left(\left(\hat{\theta} - \theta\right)^2\right) \\ &= \int \left(\hat{\theta} - \theta\right)^2 p(\mathbf{x}; \theta) d\mathbf{x}\end{aligned}$$

→ 往往不可实现

贝叶斯MSE:

$$\begin{aligned}\text{Bmse}(\hat{\theta}) &= E\left(\left(\theta - \hat{\theta}\right)^2\right) \\ &= \iint \left(\theta - \hat{\theta}\right)^2 p(\mathbf{x}, \theta) d\mathbf{x} d\theta \\ &= \iint \left(\theta - \hat{\theta}\right)^2 p(\mathbf{x}|\theta) p(\theta) d\mathbf{x} d\theta \\ &= \iint \left(\theta - \hat{\theta}\right)^2 p(\mathbf{x}|\theta) d\mathbf{x} p(\theta) d\theta\end{aligned}$$

→ $\hat{\theta} = E(\theta | \mathbf{x})$

● 存在多余参数时

多余参数：未知，但不感兴趣的参数

解决思路：通过**积分**消除多余参数的影响

(1) 后验概率中存在多余参数时：

$$p(\theta, \alpha | \mathbf{x}) \quad \rightarrow \quad p(\theta | \mathbf{x}) = \int p(\theta, \alpha | \mathbf{x}) d\alpha$$

(2) 条件概率中存在多余参数时：

$$p(\theta | \mathbf{x}) = \frac{p(\mathbf{x} | \theta) p(\theta)}{\int p(\mathbf{x} | \theta) p(\theta) d\theta}$$

若现只有 $p(\mathbf{x} | \theta, \alpha)$ ，而无 $p(\mathbf{x} | \theta)$

此时可通过**积分**方式解决

$$p(\mathbf{x} | \theta) = \int p(\mathbf{x} | \theta, \alpha) p(\alpha) d\alpha$$

进一步地，若待估计参数与多余参数相互独立，则有

$$p(\mathbf{x} | \theta) = \int p(\mathbf{x} | \theta, \alpha) p(\alpha) d\alpha$$

——充分体现了贝叶斯原理“**平均**”的思想

- 多余参数虽可通过积分方式消除，但会影响后验概率，进而影响后续估计
- 因此，在贝叶斯框架下“**好求但并不意味一定好用**”——需结合实际问题具体分析

● 矢量参数下贝叶斯估计

若 $\boldsymbol{\theta}$ 是 $p \times 1$ 的矢量参数，那么为了估计其中某个参数 θ_i ，可以将剩余参数当作**多余参数**，因此对 θ_i 的MMSE为

$$\hat{\theta}_i = E(\theta_i | \mathbf{x}) = \int \theta_i p(\theta_i | \mathbf{x}) d\theta_i$$

其中

$$p(\theta_i | \mathbf{x}) = \int \cdots \int p(\boldsymbol{\theta} | \mathbf{x}) d\theta_1 \cdots d\theta_{i-1} d\theta_{i+1} \cdots d\theta_p$$

$$\hat{\theta}_i = \int \theta_i \left(\int \cdots \int p(\boldsymbol{\theta} | \mathbf{x}) d\theta_1 \cdots d\theta_{i-1} d\theta_{i+1} \cdots d\theta_p \right) d\theta_i = \underbrace{\int \theta_i p(\boldsymbol{\theta} | \mathbf{x}) d\boldsymbol{\theta}}_{p\text{重积分}}$$

$$\hat{\boldsymbol{\theta}} = \begin{bmatrix} \int \theta_1 p(\boldsymbol{\theta} | \mathbf{x}) d\boldsymbol{\theta} \\ \int \theta_2 p(\boldsymbol{\theta} | \mathbf{x}) d\boldsymbol{\theta} \\ \vdots \\ \int \theta_p p(\boldsymbol{\theta} | \mathbf{x}) d\boldsymbol{\theta} \end{bmatrix} = \int \boldsymbol{\theta} p(\boldsymbol{\theta} | \mathbf{x}) d\boldsymbol{\theta} = E(\boldsymbol{\theta} | \mathbf{x})$$

二、贝叶斯风险

- 回顾：贝叶斯均方误差（BMSE）：

$$\min \left\{ \iint (\theta - \hat{\theta})^2 p(\mathbf{x}, \theta) d\mathbf{x} d\theta \right\}$$

误差： $\varepsilon = \theta - \hat{\theta}$ 代价函数： $C(\varepsilon) = (\theta - \hat{\theta})^2$

- 除了采用二次型误差作为代价函数外，能否采用其它代价函数？

绝对误差： $|\theta - \hat{\theta}|$ 成功失败型误差： $\begin{cases} 0, & |\theta - \hat{\theta}| < \delta \\ 1, & |\theta - \hat{\theta}| \geq \delta \end{cases} \dots$

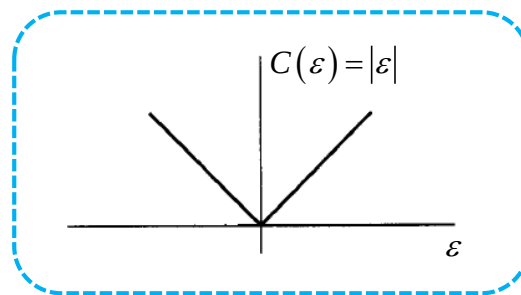
- 贝叶斯风险：

$$\mathfrak{R} = \iint C(\varepsilon) p(\mathbf{x}, \theta) d\mathbf{x} d\theta$$



• 绝对误差

绝对误差: $C(\varepsilon) = |\theta - \hat{\theta}|$



贝叶斯风险为: $\mathfrak{R} = \iint C(\varepsilon) p(\mathbf{x}, \theta) d\mathbf{x} d\theta$

$$= \int \int C(\varepsilon) p(\theta | \mathbf{x}) d\theta p(\mathbf{x}) d\mathbf{x}$$

$$\min \{\mathfrak{R}\} \longleftrightarrow \min \left\{ \int C(\varepsilon) p(\theta | \mathbf{x}) d\theta \right\}$$

$$\longrightarrow \min \left\{ \int |\theta - \hat{\theta}| p(\theta | \mathbf{x}) d\theta \right\}$$

$$\text{令 } g(\hat{\theta}) = \int |\theta - \hat{\theta}| p(\theta | \mathbf{x}) d\theta$$



$$g(\hat{\theta}) = \int |\theta - \hat{\theta}| p(\theta | \mathbf{x}) d\theta$$

$$= \int_{-\infty}^{\hat{\theta}} (\hat{\theta} - \theta) p(\theta | \mathbf{x}) d\theta + \int_{\hat{\theta}}^{+\infty} (\theta - \hat{\theta}) p(\theta | \mathbf{x}) d\theta$$

Leibnitz准则:

$$\frac{\partial}{\partial u} \int_{\phi_1(u)}^{\phi_2(u)} h(u, v) dv = \int_{\phi_1(u)}^{\phi_2(u)} \frac{\partial h(u, v)}{\partial u} dv + \frac{\partial \phi_2(u)}{\partial u} h(u, \phi_2(u)) - \frac{\partial \phi_1(u)}{\partial u} h(u, \phi_1(u))$$

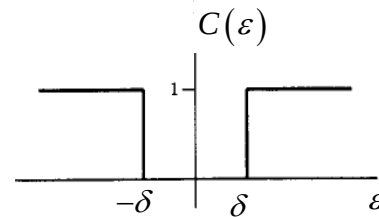
$$\begin{aligned} \rightarrow \frac{\partial g(\hat{\theta})}{\partial \hat{\theta}} &= \int_{-\infty}^{\hat{\theta}} p(\theta | \mathbf{x}) d\theta - \int_{\hat{\theta}}^{+\infty} p(\theta | \mathbf{x}) d\theta \\ \frac{\partial g(\hat{\theta})}{\partial \hat{\theta}} &= 0 \end{aligned} \quad \left. \begin{array}{l} \\ \end{array} \right\} \rightarrow \int_{-\infty}^{\hat{\theta}} p(\theta | \mathbf{x}) d\theta = \int_{\hat{\theta}}^{+\infty} p(\theta | \mathbf{x}) d\theta$$

即 $\hat{\theta}$ 是后验PDF的**中值**



● 成功失败型误差

$$\text{成功失败型误差: } C(\varepsilon) = \begin{cases} 0, & |\theta - \hat{\theta}| < \delta \\ 1, & |\theta - \hat{\theta}| \geq \delta \end{cases}$$



$$\begin{aligned} \text{贝叶斯风险为: } \mathfrak{R} &= \iint C(\varepsilon) p(\mathbf{x}, \theta) d\mathbf{x} d\theta \\ &= \int \int C(\varepsilon) p(\theta | \mathbf{x}) d\theta p(\mathbf{x}) d\mathbf{x} \end{aligned}$$

$$\min \{\mathfrak{R}\} \longleftrightarrow \min \left\{ \int C(\varepsilon) p(\theta | \mathbf{x}) d\theta \right\}$$

$$\int C(\varepsilon) p(\theta | \mathbf{x}) d\theta = \int_{-\infty}^{\hat{\theta}-\delta} p(\theta | \mathbf{x}) d\theta + \int_{\hat{\theta}+\delta}^{+\infty} p(\theta | \mathbf{x}) d\theta = 1 - \int_{\hat{\theta}-\delta}^{\hat{\theta}+\delta} p(\theta | \mathbf{x}) d\theta$$

$$\min \left\{ \int C(\varepsilon) p(\theta | \mathbf{x}) d\theta \right\} \longrightarrow \hat{\theta} = \arg \max_{\theta} p(\theta | \mathbf{x})$$

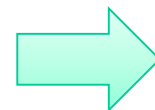
即 $\hat{\theta}$ 是后验PDF的**最大值** (众数)

MAP, maximum a posteriori



MAP: $\hat{\theta} = \arg \max_{\theta} p(\theta | \mathbf{x})$

贝叶斯公式: $p(\theta | \mathbf{x}) = \frac{p(\mathbf{x} | \theta) p(\theta)}{\int p(\mathbf{x} | \theta) p(\theta) d\theta}$



$$\hat{\theta} = \arg \max_{\theta} \{ p(\mathbf{x} | \theta) p(\theta) \}$$



$$\hat{\theta} = \arg \max_{\theta} \{ \ln p(\mathbf{x} | \theta) + \ln p(\theta) \}$$

- ✓ 相比MMSE, 无需求积分, 只需求最大值
- ✓ 可简化运算



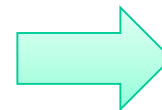
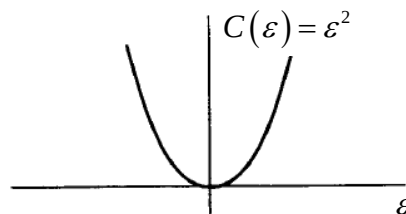
$$\text{贝叶斯风险: } \mathfrak{R} = \iint C(\varepsilon) p(\mathbf{x}, \theta) d\mathbf{x} d\theta$$

优化目标: $\min \{\mathfrak{R}\}$



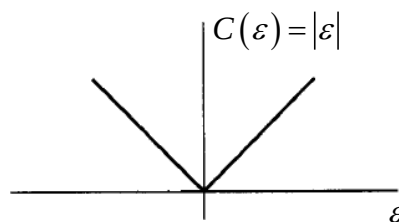
常见的代价函数

$C(\varepsilon)$ { $(\theta - \hat{\theta})^2$ 二次型误差



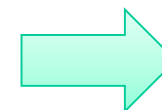
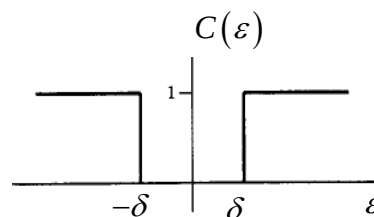
后验均值
MMSE

$C(\varepsilon)$ { $|\theta - \hat{\theta}|$ 绝对误差



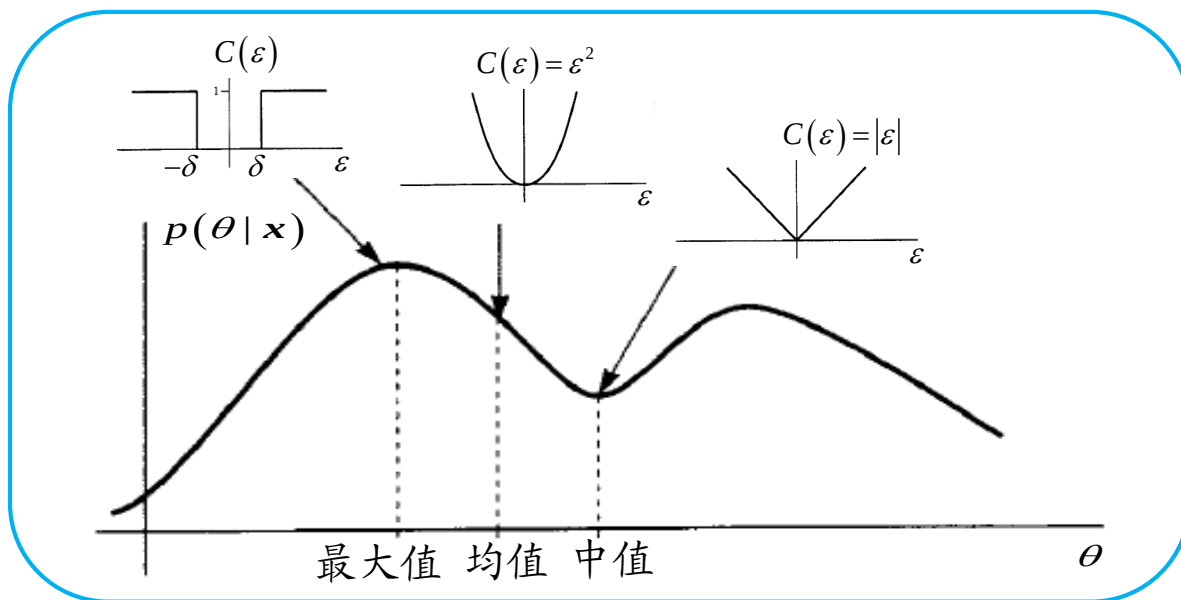
后验中值

$C(\varepsilon)$ { $\begin{cases} 0, & |\theta - \hat{\theta}| < \delta \\ 1, & |\theta - \hat{\theta}| \geq \delta \end{cases}$ 成功失败型误差



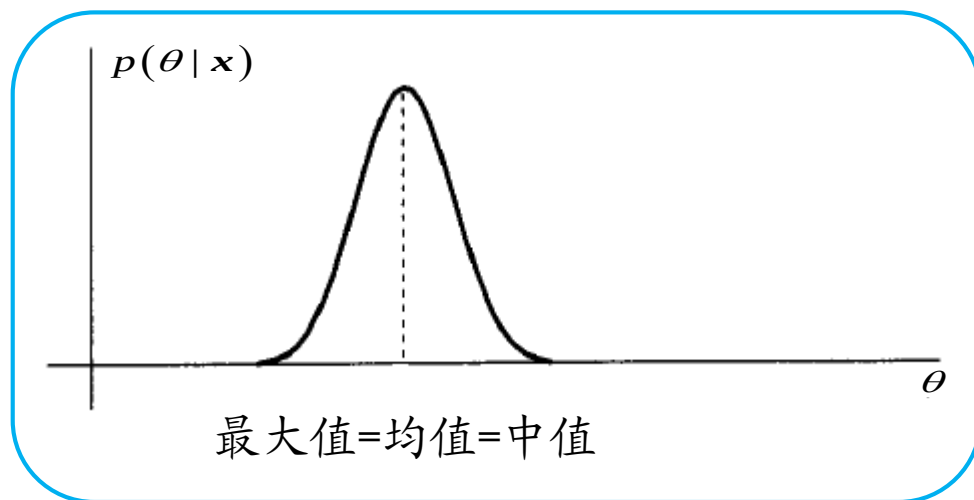
后验最大值
MAP

● 三种估计量比较



✓ 一般而言，“三值”并不相等，因此三种估计量往往不同

✓ **特例：**高斯时“三值”相等，三种估计方法等价



例：指数PDF

$$p(x[n]|\theta) = \begin{cases} \theta \exp\{-\theta x[n]\}, & x[n] > 0 \\ 0, & x[n] \leq 0 \end{cases}$$

其中 $x[n]$ 是条件独立同分布的，待估计参数为 θ ，且服从

$$p(\theta) = \begin{cases} \lambda \exp\{-\lambda \theta\}, & \theta > 0 \\ 0, & \theta \leq 0 \end{cases}$$

求 θ 的MAP估计量。

$$\text{MAP: } \hat{\theta} = \arg \max_{\theta} \{\ln p(\mathbf{x}|\theta) + \ln p(\theta)\}$$

$$p(\mathbf{x}|\theta) = \prod_{n=0}^{N-1} p(x[n]|\theta) = \begin{cases} \theta^N \exp\left\{-\theta \sum_{n=0}^{N-1} x[n]\right\}, & x[n] > 0 \\ 0, & x[n] \leq 0 \end{cases}$$

$$p(\theta) = \begin{cases} \lambda \exp\{-\lambda \theta\}, & \theta > 0 \\ 0, & \theta \leq 0 \end{cases}$$


$$\ln p(\mathbf{x}|\theta) + \ln p(\theta) = \ln \left\{ \theta^N \exp\left\{-\theta \sum_{n=0}^{N-1} x[n]\right\} \right\} + \ln \{ \lambda \exp\{-\lambda \theta\} \}$$



$$\ln p(\mathbf{x} | \theta) + \ln p(\theta) = \ln \left\{ \theta^N \exp \left\{ -\theta \sum_{n=0}^{N-1} x[n] \right\} \right\} + \ln \{ \lambda \exp \{ -\lambda \theta \} \}$$

$$= N \ln \theta - \theta N \bar{x} + \ln \lambda - \lambda \theta$$

$$\frac{\partial \{ \ln p(\mathbf{x} | \theta) + \ln p(\theta) \}}{\partial \theta} = 0 \quad \Rightarrow \quad \hat{\theta} = \frac{1}{\bar{x} + \frac{\lambda}{N}}$$

$N \rightarrow \infty?$

贝叶斯MLE:

$$\hat{\theta} = \arg \max_{\theta} \{ \ln p(\mathbf{x} | \theta) + \ln p(\theta) \} \quad \Rightarrow \quad \hat{\theta} = \arg \max_{\theta} \{ \ln p(\mathbf{x} | \theta) \}$$

- ✓ 大数据量时先验信息不起作用，最大后验概率估计（MAP）将转变为（贝叶斯）最大似然估计（MLE）

三、线性贝叶斯估计

- 观察数据 $\{x[0], x[1], x[2], \dots, x[N-1]\}$
- 线性贝叶斯估计（LMMSE，也称线性最小均方误差估计）意味着：

$$\hat{\theta} = \sum_{n=0}^{N-1} a_n x[n] + a_N$$

即限定估计量与观察数据间呈线性关系，然后使

$$\text{Bmse}(\hat{\theta}) = E\left[(\theta - \hat{\theta})^2\right] \quad \text{最小化}$$

即，LMMSE:

$$\begin{cases} \min \left\{ E\left[(\theta - \hat{\theta})^2\right] \right\} \\ \text{s.t. } \hat{\theta} = \sum_{n=0}^{N-1} a_n x[n] + a_N \end{cases}$$

LMMSE:

$$\begin{cases} \min \left\{ E \left[\left(\theta - \hat{\theta} \right)^2 \right] \right\} \\ \text{s.t. } \hat{\theta} = \sum_{n=0}^{N-1} a_n x[n] + a_N \end{cases}$$

$$\left. \begin{aligned} \text{Bmse}(\hat{\theta}) &= E \left[\left(\theta - \hat{\theta} \right)^2 \right] \\ \hat{\theta} &= \sum_{n=0}^{N-1} a_n x[n] + a_N \end{aligned} \right\} \Rightarrow \left. \begin{aligned} \text{Bmse}(\hat{\theta}) &= E \left[\left(\theta - \sum_{n=0}^{N-1} a_n x[n] - a_N \right)^2 \right] \\ \frac{\partial \text{Bmse}(\hat{\theta})}{\partial a_N} &= 0 \end{aligned} \right\} \Rightarrow$$

$$E \left[\left(\theta - \sum_{n=0}^{N-1} a_n x[n] - a_N \right) \right] = 0 \Rightarrow a_N = E(\theta) - \sum_{n=0}^{N-1} a_n E(x[n])$$



$$\left. \begin{aligned} a_N &= E(\theta) - \sum_{n=0}^{N-1} a_n E(x[n]) \\ \hat{\theta} &= \sum_{n=0}^{N-1} a_n x[n] + a_N \end{aligned} \right\} \Rightarrow \left. \begin{aligned} \hat{\theta} &= E(\theta) + \sum_{n=0}^{N-1} a_n (x[n] - E(x[n])) \\ \text{Bmse}(\hat{\theta}) &= E[(\theta - \hat{\theta})^2] \end{aligned} \right\} \Rightarrow$$

$$\begin{aligned} \text{Bmse}(\hat{\theta}) &= E \left[\left(\theta - E(\theta) - \sum_{n=0}^{N-1} a_n (x[n] - E(x[n])) \right)^2 \right] \\ &= E \left[\left(\underbrace{\sum_{n=0}^{N-1} a_n (x[n] - E(x[n]))}_{\mathbf{a} = [a_0, a_1, a_2, \dots, a_{N-1}]^T} - (\theta - E(\theta)) \right)^2 \right]
 \end{aligned}$$

$$= E \left[\left(\mathbf{a}^T (\mathbf{x} - E(\mathbf{x})) - (\theta - E(\theta)) \right)^2 \right]$$

$$= E \left[\left(\mathbf{a}^T (\mathbf{x} - E(\mathbf{x})) - (\theta - E(\theta)) \right) \left(\mathbf{a}^T (\mathbf{x} - E(\mathbf{x})) - (\theta - E(\theta)) \right)^T \right]$$



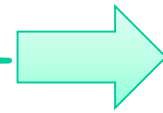
$$\text{Bmse}(\hat{\theta}) = E \left[\left(\mathbf{a}^T (\mathbf{x} - E(\mathbf{x})) - (\theta - E(\theta)) \right) \left(\mathbf{a}^T (\mathbf{x} - E(\mathbf{x})) - (\theta - E(\theta)) \right)^T \right]$$

$$= E \left[\begin{aligned} &\mathbf{a}^T (\mathbf{x} - E(\mathbf{x})) (\mathbf{x} - E(\mathbf{x}))^T \mathbf{a} - \mathbf{a}^T (\mathbf{x} - E(\mathbf{x})) (\theta - E(\theta)) \\ &- (\theta - E(\theta)) (\mathbf{x} - E(\mathbf{x}))^T \mathbf{a} + (\theta - E(\theta))^2 \end{aligned} \right]$$

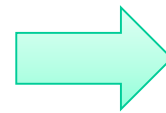
$$\mathbf{C}_{\theta x}^T = \mathbf{C}_{x\theta}$$

$$= \mathbf{a}^T \mathbf{C}_{xx} \mathbf{a} - \mathbf{a}^T \mathbf{C}_{x\theta} - \mathbf{C}_{\theta x} \mathbf{a} + \mathbf{C}_{\theta\theta}$$

$$\frac{\partial \text{Bmse}(\hat{\theta})}{\partial \mathbf{a}} = 0$$



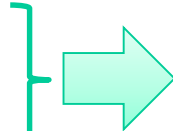
$$2\mathbf{C}_{xx} \mathbf{a} - 2\mathbf{C}_{x\theta} = 0$$



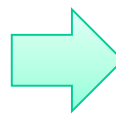
$$\mathbf{a} = \mathbf{C}_{xx}^{-1} \mathbf{C}_{x\theta}$$

$$\hat{\theta} = \sum_{n=0}^{N-1} a_n x[n] + a_N = \mathbf{a}^T \mathbf{x} + a_N$$

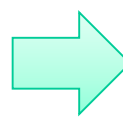
$$a_N = E(\theta) - \sum_{n=0}^{N-1} a_n E(x[n]) = E(\theta) - \mathbf{a}^T E(\mathbf{x})$$



$$\hat{\theta} = E(\theta) + \mathbf{a}^T (\mathbf{x} - E(\mathbf{x}))$$



$$\hat{\theta} = E(\theta) + \mathbf{C}_{\theta x} \mathbf{C}_{xx}^{-1} (\mathbf{x} - E(\mathbf{x}))$$



$$\text{Bmse}(\hat{\theta}) = \mathbf{C}_{\theta\theta} - \mathbf{C}_{\theta x} \mathbf{C}_{xx}^{-1} \mathbf{C}_{x\theta}$$

$$\text{Bmse}(\hat{\theta}) = E \left[(\theta - \hat{\theta})^2 \right]$$

- ✓ 仅需一阶矩、二阶矩
- ✓ 无需PDF



● LMMSE Vs MMSE

LMMSE

LMMSE优化目标

$$\begin{cases} \min E \left[(\theta - \hat{\theta})^2 \right] \\ \text{s.t. } \hat{\theta} = \sum_{n=0}^{N-1} a_n x[n] + a_N \end{cases}$$

相应的LMMSE估计量及Bmse

$$\begin{cases} \hat{\theta} = E(\theta) + \mathbf{C}_{\theta x} \mathbf{C}_{xx}^{-1} (\mathbf{x} - E(\mathbf{x})) \\ \text{Bmse}(\hat{\theta}) = \mathbf{C}_{\theta\theta} - \mathbf{C}_{\theta x} \mathbf{C}_{xx}^{-1} \mathbf{C}_{x\theta} \end{cases}$$

- ✓ 附加了**线性约束**
- ✓ 仅需一阶矩和二阶矩
- ✓ 可得显示解——好求
- ✓ 仅在“线性”中最优

MMSE

MMSE优化目标: $\min \left\{ E \left((\theta - \hat{\theta})^2 \right) \right\}$

相应的MMSE估计量及Bmse

$$\begin{cases} \hat{\theta} = E(\theta | \mathbf{x}) \\ \text{Bmse}(\hat{\theta}) = E \left((\theta - \hat{\theta})^2 \right) \end{cases}$$

- ✓ 无附加约束
- ✓ 需PDF
- ✓ 可能难以求得显示解
- ✓ 全局最优

● 多维高斯条件PDF定理

若 $\mathbf{x}$ 和 $\mathbf{y}$ 是联合高斯，其中 $\mathbf{x}$ 是 $k \times 1$ 维， $\mathbf{y}$ 是 $l \times 1$ 维，均值矢量是 $\begin{bmatrix} E(\mathbf{x})^T, E(\mathbf{y})^T \end{bmatrix}^T$ ，分块协方差矩阵是

$$\mathbf{C} = \begin{bmatrix} \mathbf{C}_{xx} & \mathbf{C}_{xy} \\ \mathbf{C}_{yx} & \mathbf{C}_{yy} \end{bmatrix}$$

联合概率密度函数是

$$p(\mathbf{x}, \mathbf{y}) = \frac{1}{(2\pi)^{\frac{k+l}{2}} (\det(\mathbf{C}))^{\frac{1}{2}}} \exp \left[-\frac{1}{2} \begin{pmatrix} \mathbf{x} - E(\mathbf{x}) \\ \mathbf{y} - E(\mathbf{y}) \end{pmatrix}^T \mathbf{C}^{-1} \begin{pmatrix} \mathbf{x} - E(\mathbf{x}) \\ \mathbf{y} - E(\mathbf{y}) \end{pmatrix} \right]$$

那么条件概率也是高斯的，且

$$E(\mathbf{y} | \mathbf{x}) = E(\mathbf{y}) + \mathbf{C}_{yx} \mathbf{C}_{xx}^{-1} (\mathbf{x} - E(\mathbf{x}))$$

$$\mathbf{C}_{y|x} = \mathbf{C}_{yy} - \mathbf{C}_{yx} \mathbf{C}_{xx}^{-1} \mathbf{C}_{xy}$$



若观测数据 $\mathbf{x}$ 满足**贝叶斯一般线性模型**:

$$\mathbf{x} = \mathbf{H}\boldsymbol{\theta} + \mathbf{w}$$

其中, $\mathbf{H}$ 为观测矩阵, $\boldsymbol{\theta}$ 为待估计参数且服从高斯分布 $N(\boldsymbol{\mu}_\theta, \mathbf{C}_\theta)$ 噪声矢量 $\mathbf{w}$ 服从高斯分布 $N(\mathbf{0}, \mathbf{C}_w)$ 且与 $\boldsymbol{\theta}$ 无关, 则MMSE为:

$$\hat{\boldsymbol{\theta}} = E(\boldsymbol{\theta} | \mathbf{x}) = E(\boldsymbol{\theta}) + \mathbf{C}_{\theta\mathbf{x}} \mathbf{C}_{\mathbf{x}\mathbf{x}}^{-1} (\mathbf{x} - E(\mathbf{x}))$$

$$\mathbf{C}_{\theta|\mathbf{x}} = \mathbf{C}_{\theta\theta} - \mathbf{C}_{\theta\mathbf{x}} \mathbf{C}_{\mathbf{x}\mathbf{x}}^{-1} \mathbf{C}_{\mathbf{x}\theta}$$

Vs

LMMSE

$$\hat{\boldsymbol{\theta}} = E(\boldsymbol{\theta}) + \mathbf{C}_{\theta\mathbf{x}} \mathbf{C}_{\mathbf{x}\mathbf{x}}^{-1} (\mathbf{x} - E(\mathbf{x}))$$

$$\text{Bmse}(\hat{\boldsymbol{\theta}}) = \mathbf{C}_{\theta\theta} - \mathbf{C}_{\theta\mathbf{x}} \mathbf{C}_{\mathbf{x}\mathbf{x}}^{-1} \mathbf{C}_{\mathbf{x}\theta}$$

对贝叶斯一般线性模型, LMMSE与MMSE具有相同形式

例：白噪声中电平估计问题：

$$x[n] = A + w[n], \quad n = 0, 1, \dots, N-1$$

待估计参数为信号幅度 A ，且 $A \sim U[-A_0, A_0]$ ， $w[n]$ 为高斯白噪声 $w[n] \sim N(0, \sigma^2)$ ，且和 A 相互独立。试求MMSE、MAP与LMMSE

1. MMSE估计

$$\hat{A} = \int \underline{A p(A | \mathbf{x})} dA$$

$$p(A | \mathbf{x}) = \frac{p(\mathbf{x} | A) p(A)}{\int p(\mathbf{x} | A) p(A) dA}$$

$$p_x(x[n] | A) = p_w(x[n] - A | A)$$

$$= p_w(x[n] - A)$$

$$= \frac{1}{\sqrt{2\pi\sigma^2}} \exp\left\{-\frac{1}{2\sigma^2}(x[n] - A)^2\right\}$$

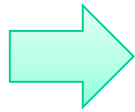


$$\begin{aligned}
 & \rightarrow p(\mathbf{x} | A) = \frac{1}{(2\pi\sigma^2)^{N/2}} \exp \left\{ -\frac{1}{2\sigma^2} \sum_{n=0}^{N-1} (x[n] - A)^2 \right\} \\
 & p(A) = \frac{1}{2A_0}, \quad |A| \leq A_0 \\
 & p(A | \mathbf{x}) = \frac{p(\mathbf{x} | A) p(A)}{\int p(\mathbf{x} | A) p(A) dA}
 \end{aligned}
 \quad \left. \vphantom{\begin{aligned} p(\mathbf{x} | A) \\ p(A) \\ p(A | \mathbf{x}) \end{aligned}} \right\} \rightarrow$$

$$p(A | \mathbf{x}) = \begin{cases} \frac{\frac{1}{2A_0 (2\pi\sigma^2)^{N/2}} \exp \left\{ -\frac{1}{2\sigma^2} \sum_{n=0}^{N-1} (x[n] - A)^2 \right\}}{\int_{-A_0}^{A_0} \frac{1}{2A_0 (2\pi\sigma^2)^{N/2}} \exp \left\{ -\frac{1}{2\sigma^2} \sum_{n=0}^{N-1} (x[n] - A)^2 \right\} dA}, & |A| \leq A_0 \\ 0, & |A| > A_0 \end{cases}$$

$$= \begin{cases} \frac{\exp \left\{ -\frac{1}{2\sigma^2} N (A - \bar{x})^2 \right\}}{\int_{-A_0}^{A_0} \exp \left\{ -\frac{1}{2\sigma^2} N (A - \bar{x})^2 \right\} dA}, & |A| \leq A_0 \\ 0, & |A| > A_0 \end{cases}$$





$$p(A|\mathbf{x}) = \begin{cases} \frac{\exp\left\{-\frac{1}{2\frac{\sigma^2}{N}}(A-\bar{x})^2\right\}}{\int_{-A_0}^{A_0} \exp\left\{-\frac{1}{2\frac{\sigma^2}{N}}(A-\bar{x})^2\right\} dA}, & |A| \leq A_0 \\ 0, & |A| > A_0 \end{cases} = \begin{cases} \frac{\frac{1}{\sqrt{2\pi\frac{\sigma^2}{N}}} \exp\left\{-\frac{1}{2\frac{\sigma^2}{N}}(A-\bar{x})^2\right\}}{\int_{-A_0}^{A_0} \frac{1}{\sqrt{2\pi\frac{\sigma^2}{N}}} \exp\left\{-\frac{1}{2\frac{\sigma^2}{N}}(A-\bar{x})^2\right\} dA}, & |A| \leq A_0 \\ 0, & |A| > A_0 \end{cases}$$

$$\text{MMSE: } \hat{A} = E(A|\mathbf{x}) = \int A p(A|\mathbf{x}) dA$$

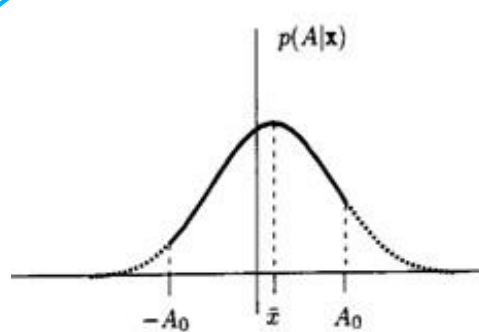
$$= \frac{\int_{-A_0}^{A_0} A \frac{1}{\sqrt{2\pi\frac{\sigma^2}{N}}} \exp\left\{-\frac{1}{2\frac{\sigma^2}{N}} \sum_{n=0}^{N-1} (A-\bar{x})^2\right\} dA}{\int_{-A_0}^{A_0} \frac{1}{\sqrt{2\pi\frac{\sigma^2}{N}}} \exp\left\{-\frac{1}{2\frac{\sigma^2}{N}} \sum_{n=0}^{N-1} (A-\bar{x})^2\right\} dA}$$

——化简困难！



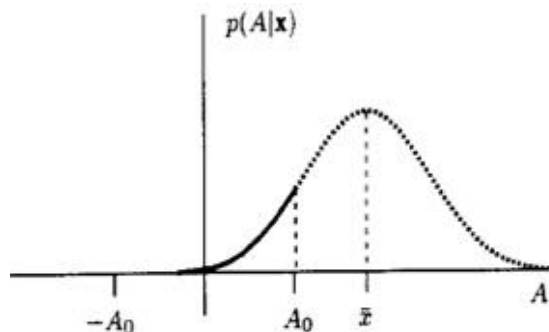
2. MAP估计?

$$p(A|\mathbf{x}) = \begin{cases} \frac{1}{\sqrt{2\pi\frac{\sigma^2}{N}}} \exp\left\{-\frac{1}{2\frac{\sigma^2}{N}}(A-\bar{x})^2\right\} \\ \int_{-A_0}^{A_0} \frac{1}{\sqrt{2\pi\frac{\sigma^2}{N}}} \exp\left\{-\frac{1}{2\frac{\sigma^2}{N}}(A-\bar{x})^2\right\} dA \\ 0, \end{cases}, \quad \begin{matrix} |A| \leq A_0 \\ |A| > A_0 \end{matrix}$$



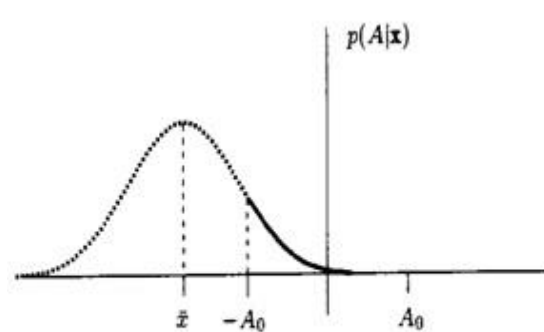
(a) $-A_0 \leq \bar{x} \leq A_0$

$\bar{x}$



(b) $\bar{x} > A_0$

A_0



(c) $\bar{x} < -A_0$

$-A_0$

$$\tilde{A} = \begin{cases} -A_0, & \bar{x} < -A_0 \\ \bar{x}, & -A_0 \leq \bar{x} \leq A_0 \\ A_0, & \bar{x} > A_0 \end{cases}$$



3. LMMSE估计量

$$\hat{A} = E(A) + \mathbf{C}_{Ax} \mathbf{C}_{xx}^{-1} (\mathbf{x} - E(\mathbf{x}))$$

其中, $E(A) = 0$

$$\mathbf{C}_{Ax} = E\left((A - E(A))(\mathbf{x} - E(\mathbf{x}))^T\right) = E\left(A(\mathbf{A}\mathbf{1} + \mathbf{w})^T\right) = E(A^2)\mathbf{1}^T = \sigma_A^2 \mathbf{1}^T$$

$$\mathbf{C}_{xx} = E\left((\mathbf{x} - E(\mathbf{x}))(\mathbf{x} - E(\mathbf{x}))^T\right) = E\left((\mathbf{A}\mathbf{1} + \mathbf{w})(\mathbf{A}\mathbf{1} + \mathbf{w})^T\right) = \sigma_A^2 \mathbf{1}\mathbf{1}^T + \sigma^2 \mathbf{I}$$

$$\hat{A} = \sigma_A^2 \mathbf{1}^T (\sigma_A^2 \mathbf{1}\mathbf{1}^T + \sigma^2 \mathbf{I})^{-1} \mathbf{x}$$

$$= \frac{\sigma_A^2}{\sigma^2} \mathbf{1}^T \left(\frac{\sigma_A^2}{\sigma^2} \mathbf{1}\mathbf{1}^T + \mathbf{I} \right)^{-1} \mathbf{x}$$

Woodbury恒等式:

$$(\mathbf{B} + \mathbf{u}\mathbf{u}^T)^{-1} = \mathbf{B}^{-1} - \frac{\mathbf{B}^{-1}\mathbf{u}\mathbf{u}^T\mathbf{B}^{-1}}{1 + \mathbf{u}^T\mathbf{B}^{-1}\mathbf{u}}$$

$$\hat{A} = \frac{\sigma_A^2}{\sigma^2} \mathbf{1}^T \left(\mathbf{I} - \frac{\frac{\sigma_A^2}{\sigma^2} \mathbf{1}\mathbf{1}^T}{1 + \frac{\sigma_A^2}{\sigma^2} N} \right) \mathbf{x}$$
$$= \frac{\sigma_A^2}{\sigma_A^2 + \frac{\sigma^2}{N}} \bar{x} \quad \text{——显示解}$$

✓ 比MMSE更好求解!

✓ 但, 是**准最佳的!**

● 推广至矢量参数情况

待估计参数 $\boldsymbol{\theta} = [\theta_1, \theta_2, \dots, \theta_p]^T$ ，其每个参数的LMMSE定义为：

$$\begin{cases} \min E \left[\left(\theta_i - \hat{\theta}_i \right)^2 \right] \\ \text{s.t. } \hat{\theta}_i = \sum_{n=0}^{N-1} a_{in} x[n] + a_{iN} \end{cases}$$

标量参数LMMSE：

$$\hat{\theta} = E(\theta) + \mathbf{C}_{\theta x} \mathbf{C}_{xx}^{-1} (\mathbf{x} - E(\mathbf{x}))$$

$$\text{Bmse}(\hat{\theta}) = C_{\theta\theta} - \mathbf{C}_{\theta x} \mathbf{C}_{xx}^{-1} \mathbf{C}_{x\theta}$$



$$\begin{bmatrix} \hat{\theta}_1 \\ \hat{\theta}_2 \\ \vdots \\ \hat{\theta}_p \end{bmatrix} = \begin{bmatrix} E(\theta_1) \\ E(\theta_2) \\ \vdots \\ E(\theta_p) \end{bmatrix} + \begin{bmatrix} \mathbf{C}_{\theta_1 x} \mathbf{C}_{xx}^{-1} (\mathbf{x} - E(\mathbf{x})) \\ \mathbf{C}_{\theta_2 x} \mathbf{C}_{xx}^{-1} (\mathbf{x} - E(\mathbf{x})) \\ \vdots \\ \mathbf{C}_{\theta_p x} \mathbf{C}_{xx}^{-1} (\mathbf{x} - E(\mathbf{x})) \end{bmatrix}$$

$$= E(\boldsymbol{\theta}) + \mathbf{C}_{\theta x} \mathbf{C}_{xx}^{-1} (\mathbf{x} - E(\mathbf{x}))$$

—— 矢量参数LMMSE



四、序贯LMMSE

例：白噪声中电平估计问题：

$$x[n] = A + w[n], \quad n = 0, 1, \dots, N-1$$

待估计参数为信号幅度 A ，其服从 $N(0, \sigma_A^2)$ ， $w[n]$ 为高斯白噪声，且其方差为 σ^2 ，即 $w[n] \sim N(0, \sigma^2)$ ，且与幅度不相关。 A 的LMMSE估计量为？

$$\hat{\theta} = E(\theta) + \mathbf{C}_{\theta x} \mathbf{C}_{xx}^{-1} (\mathbf{x} - E(\mathbf{x}))$$

$$\text{Bmse}(\hat{\theta}) = \mathbf{C}_{\theta\theta} - \mathbf{C}_{\theta x} \mathbf{C}_{xx}^{-1} \mathbf{C}_{x\theta}$$

$$E(\theta) = 0$$

$$\mathbf{C}_{\theta x} = E\left((\theta - E(\theta))(\mathbf{x} - E(\mathbf{x}))^T\right)$$

$$= E\left((A - 0)(\mathbf{x} - 0)^T\right)$$

$$= E\left(A(\mathbf{A}\mathbf{1} + \mathbf{w})^T\right)$$

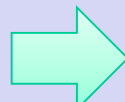
$$= \sigma_A^2 \mathbf{1}^T$$

$$\begin{aligned} \mathbf{C}_{xx} &= E\left((\mathbf{x} - E(\mathbf{x}))(\mathbf{x} - E(\mathbf{x}))^T\right) \\ &= E\left((\mathbf{A}\mathbf{1} + \mathbf{w})(\mathbf{A}\mathbf{1} + \mathbf{w})^T\right) \\ &= E\left\{\mathbf{A}\mathbf{1}\mathbf{1}^T \mathbf{A} + \mathbf{A}\mathbf{1}\mathbf{w}^T + \mathbf{w}\mathbf{1}^T \mathbf{A} + \mathbf{w}\mathbf{w}^T\right\} \\ &= \sigma_A^2 \mathbf{1}\mathbf{1}^T + \sigma^2 \mathbf{I} \end{aligned}$$

$$\mathbf{C}_{xx}^{-1} = \frac{1}{\sigma^2} \left(\mathbf{I} - \frac{\frac{\sigma_A^2}{\sigma^2} \mathbf{1}\mathbf{1}^T}{1 + \frac{\sigma_A^2}{\sigma^2} N} \right)$$

$$\mathbf{C}_{\theta\theta} = \sigma_A^2$$

$$\begin{aligned} \mathbf{C}_{x\theta} &= \mathbf{C}_{\theta x}^T \\ &= \sigma_A^2 \mathbf{1} \end{aligned}$$



$$\left\{ \begin{aligned} \hat{A} &= \frac{\sigma_A^2}{\sigma_A^2 + \frac{\sigma^2}{N}} \bar{x} \\ \text{Bmse}(\hat{A}) &= \frac{\sigma_A^2 \sigma^2}{N \sigma_A^2 + \sigma^2} \end{aligned} \right.$$

能否序贯计算？

● 对LMMSE估计量

将估计量重记为：

$$\hat{A}[N-1] = \frac{\sigma_A^2}{\sigma_A^2 + \frac{\sigma^2}{N}} \frac{1}{N} \sum_{n=0}^{N-1} x[n]$$



$$\hat{A}[N] = \frac{\sigma_A^2}{\sigma_A^2 + \frac{\sigma^2}{N+1}} \frac{1}{N+1} \sum_{n=0}^N x[n]$$

$$= \frac{N \sigma_A^2}{(N+1) \sigma_A^2 + \sigma^2} \frac{1}{N} \left(\sum_{n=0}^{N-1} x[n] + x[N] \right)$$

$$\begin{aligned} &= \frac{N \sigma_A^2}{(N+1) \sigma_A^2 + \sigma^2} \frac{1}{N} \sum_{n=0}^{N-1} x[n] + \frac{N \sigma_A^2}{(N+1) \sigma_A^2 + \sigma^2} \frac{1}{N} x[N] \\ &= \frac{N \sigma_A^2}{(N+1) \sigma_A^2 + \sigma^2} \frac{\sigma_A^2 + \frac{\sigma^2}{N}}{\sigma_A^2} \hat{A}[N-1] + \frac{\sigma_A^2}{(N+1) \sigma_A^2 + \sigma^2} x[N] \\ &= \frac{N \sigma_A^2 + \sigma^2}{(N+1) \sigma_A^2 + \sigma^2} \hat{A}[N-1] + \frac{\sigma_A^2}{(N+1) \sigma_A^2 + \sigma^2} x[N] \\ &= \hat{A}[N-1] + \frac{\sigma_A^2}{(N+1) \sigma_A^2 + \sigma^2} (x[N] - \hat{A}[N-1]) \end{aligned}$$

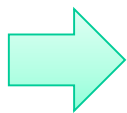
$K[N]$: 增益因子

$$\begin{aligned} &\frac{\frac{1}{\sigma^2}}{\frac{1}{\text{Bmse}(\hat{A}[N-1])} + \frac{1}{\sigma^2}} = \frac{\text{Bmse}(\hat{A}[N-1])}{\text{Bmse}(\hat{A}[N-1]) + \sigma^2} \\ &= \frac{\frac{\sigma_A^2 \sigma^2}{N \sigma_A^2 + \sigma^2}}{\frac{\sigma_A^2 \sigma^2}{N \sigma_A^2 + \sigma^2} + \sigma^2} = \frac{\sigma_A^2}{(N \sigma_A^2 + \sigma^2) + \sigma_A^2} = K[N] \end{aligned}$$

● 对贝叶斯均方误差

将Bmse重记为：

$$\text{Bmse}(\hat{A}[N-1]) = \frac{\sigma_A^2 \sigma^2}{N\sigma_A^2 + \sigma^2}$$



$$\begin{aligned}\text{Bmse}(\hat{A}[N]) &= \frac{\sigma_A^2 \sigma^2}{(N+1)\sigma_A^2 + \sigma^2} \\ &= \frac{N\sigma_A^2 + \sigma^2}{(N+1)\sigma_A^2 + \sigma^2} \frac{\sigma_A^2 \sigma^2}{N\sigma_A^2 + \sigma^2} \\ &= \left(1 - \frac{\sigma_A^2}{(N+1)\sigma_A^2 + \sigma^2}\right) \frac{\sigma_A^2 \sigma^2}{N\sigma_A^2 + \sigma^2} \\ &= (1 - K[N]) \text{Bmse}(\hat{A}[N-1])\end{aligned}$$

序贯计算方法

估计量更新：

$$\hat{A}[N] = \hat{A}[N-1] + K[N](x[N] - \hat{A}[N-1])$$

增益因子：

$$K[N] = \frac{\text{Bmse}(\hat{A}[N-1])}{\text{Bmse}(\hat{A}[N-1]) + \sigma^2}$$

最小贝叶斯MSE更新：

$$\text{Bmse}(\hat{A}[N]) = (1 - K[N]) \text{Bmse}(\hat{A}[N-1])$$

初始化：

$$\hat{A}[-1] = E(A)$$

$$\text{Bmse}(\hat{A}[-1]) = \text{var}(A)$$

五、小结

■ 贝叶斯风险

——通过引入“代价”函数，实现对一般贝叶斯估计“扩展”

- 二次型误差：后验概率的均值（MMSE估计量）
- 绝对误差：后验概率的中值
- 成功失败型误差：后验概率的最大值（MAP估计量）

■ 线性最小均方误差估计（LMMSE）

- 限定估计量是线性的，然后再找贝叶斯MSE最小者
 - 是对MMSE的“妥协”
 - 优点：无需PDF，仅需一、二阶矩即可；相比MMSE更易求解
 - 缺点：一般情况下并非全局最佳（在贝叶斯MSE准则下）

■ 序贯LMMSE估计

- 新估计量可通过对旧估计量更新的方式得到